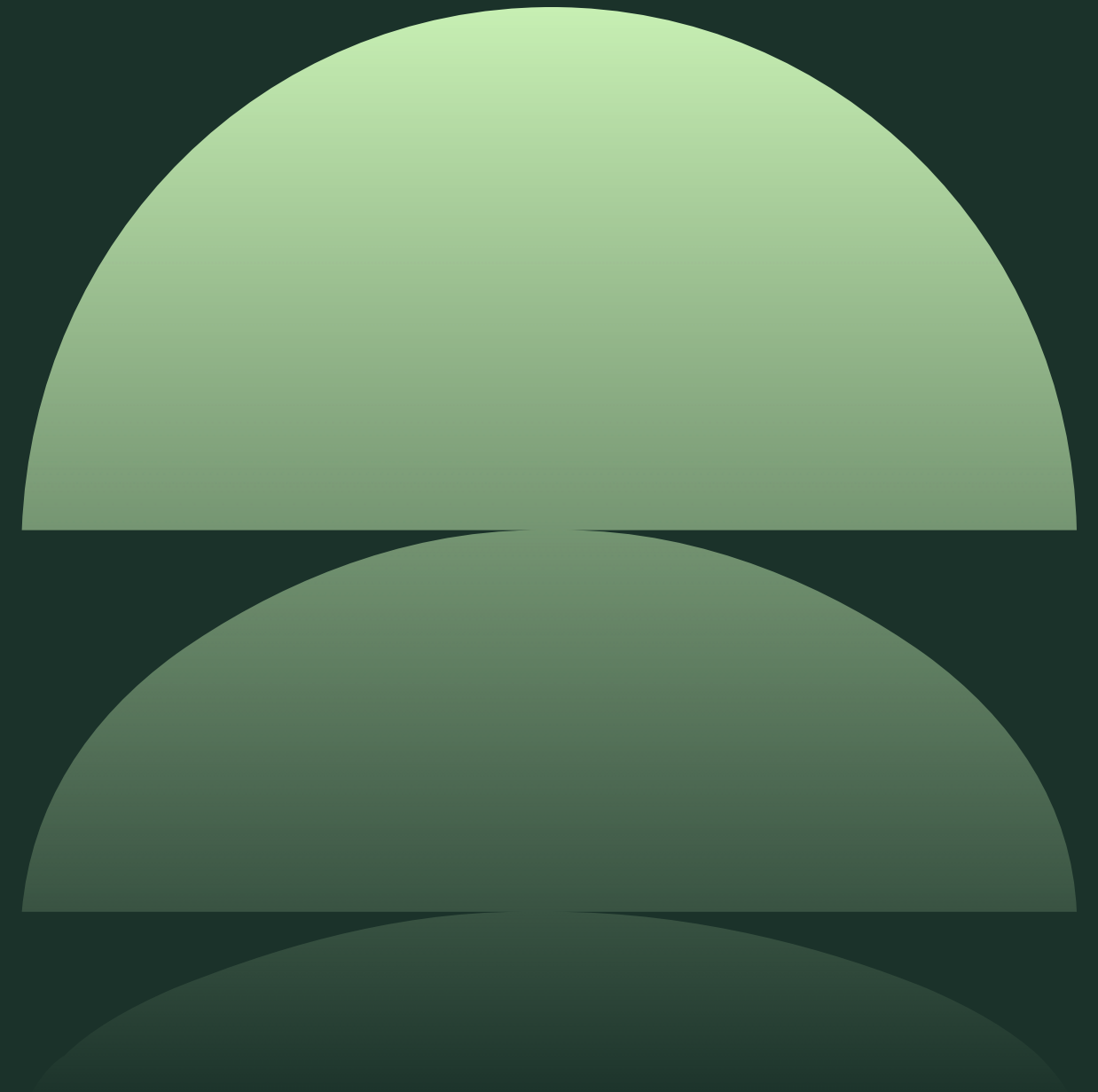
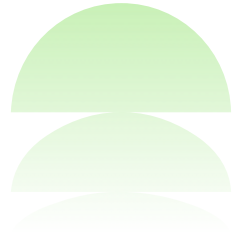


E-COMMERCE ANALYSIS REPORT



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Introduction

This project analyzes e-commerce sales data using SQL to understand customer behavior, sales performance, revenue trends, and product performance. It uses SQL queries to transform raw data into meaningful business insights.





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ANSWERS START FROM SLIDE:- 8



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What percentage does each category contribute to total revenue?

01

What is our total revenue so far?

```
SELECT
    ROUND(SUM(order_items.quantity * order_items.unit_price)) AS total_revenue
FROM
    order_items;
```

	total_revenue
▶	39553815

02

How many total orders have we received?

```
SELECT
    COUNT(orders.order_id) AS total_orders
FROM
    orders;
```

	total_orders
▶	2000

03

How many total customers do we have?

```
SELECT
    COUNT(customers.customer_id) AS total_cst
FROM
    customers;
```

	total_cst
▶	90

04

How many total products do we sell?

```
SELECT
    COUNT(products.product_id) AS total_prod
FROM
    products;
```

	total_prod
▶	38

05

What is the average amount a customer spends per order?

```
SELECT
    ROUND(SUM(quantity * unit_price) / COUNT(DISTINCT order_id),
          2) AS avg_order_value
FROM
    order_items;
```

	avg_order_value
▶	19776.91

06

What are the top 5 best-selling products?

```
SELECT
    products.product_name, SUM(order_items.quantity) AS quantity
FROM
    products
    JOIN
    order_items ON products.product_id = order_items.product_id
GROUP BY products.product_name
ORDER BY quantity DESC
LIMIT 5;
```

```
SELECT
    p.product_name, SUM(oi.quantity) AS total_qty
FROM
    order_items oi
    JOIN
    products p ON oi.product_id = p.product_id
GROUP BY p.product_name
ORDER BY total_qty DESC
LIMIT 5;
```

	product_name	quantity
▶	Smartphone X12	474
	Fiction Novel Pack	458
	Badminton Racket	457
	Cricket Bat	450
	Organic Rice 5kg	433

07

Which category is generating the most revenue?

```
SELECT
    p.category,
    ROUND(SUM(oi.quantity * oi.unit_price), 2) AS total_rv
FROM
    products AS p
    JOIN
    order_items AS oi ON p.product_id = oi.product_id
GROUP BY p.category
ORDER BY total_rv DESC;
```

	category	total_rv
▶	Electronics	25223126.00
	Home & Kitchen	5080392.00
	Clothing	4080449.00
	Sports	2064623.00
	Beauty	1829532.00
	Grocery	683745.00
	Books	591948.00

08

Who are our top 5 most valuable customers?

```
SELECT
    c.customer_name,
    ROUND(SUM(oi.quantity * oi.unit_price), 2) AS rev
FROM
    customers c
    JOIN
    orders o ON o.customer_id = c.customer_id
    JOIN
    order_items oi ON oi.order_id = o.order_id
GROUP BY c.customer_name
ORDER BY rev DESC
LIMIT 5;
```

	customer_name	rev
▶	Rahul Iyer	1673772.00
	Nikhil Mishra	1529530.00
	Riya Rana	1523531.00
	Simran Verma	1500017.00
	Manish Gupta	1449062.00

09

How has our revenue trended month by month?

```
SELECT
  DATE_FORMAT(o.order_date, '%Y-%M') AS month,
  ROUND(SUM(oi.quantity * oi.unit_price), 2) AS rev
FROM
  orders o
  JOIN
  order_items oi ON o.order_id = oi.order_id
GROUP BY month
ORDER BY month DESC;
```

	month	rev
▶	2026-May	3676986.00
	2026-March	2648708.00
	2026-June	5171310.00
	2026-July	4810007.00
	2026-January	2601033.00
	2026-February	1618245.00
	2026-April	2957803.00
	2025-September	1304388.00
	2025-October	1729314.00

10

Which city is generating the most sales?

```
SELECT
  c.city, ROUND(SUM(oi.quantity * oi.unit_price), 2) AS rev
FROM
  customers AS c
  JOIN
  orders AS o ON o.customer_id = c.customer_id
  JOIN
  order_items as oi ON oi.order_id = o.order_id
GROUP BY c.city
ORDER BY rev DESC
LIMIT 1;
```

	city	rev
▶	Delhi	5435283.00

11

Which customers purchase repeatedly?

```
SELECT
    customer_id, COUNT(order_id) AS total_orders
FROM
    orders
GROUP BY customer_id
HAVING COUNT(order_id) > 1
ORDER BY total_orders DESC;
```

```
SELECT
    c.customer_id,
    c.customer_name,
    COUNT(o.order_id) AS total_orders
FROM orders o
JOIN customers c
    ON o.customer_id = c.customer_id
GROUP BY c.customer_id, c.customer_name
HAVING COUNT(o.order_id) > 1
ORDER BY total_orders DESC;
```

	customer_id	customer_name	total_orders
▶	25	Komal Malhotra	79
	30	Nikhil Mishra	77
	74	Neha Sharma	67
	68	Pallavi Chauhan	65
	13	Rahul Iyer	65
	77	Anil Verma	64
	46	Simran Verma	62
	71	Deepak Chauhan	61
	43	Riya Rana	61
	2	Rahul Rana	58

12

Which customers have never placed an order?

```
SELECT
    c.customer_id, c.customer_name
FROM
    customers AS c
    LEFT JOIN
        orders o ON c.customer_id = o.customer_id
WHERE
    o.order_id IS NULL;
```

	customer_id	customer_name
▶	4	Vivek Chopra
	18	Nisha Joshi
	32	Nikhil Chauhan
	49	Kavita Bansal
	54	Gaurav Joshi
	58	Ritu Agarwal
	60	Gaurav Yadav
	63	Divya Joshi
	84	Neha Kapoor
	87	Deepak Nair

13

Which customers have been inactive in the last 90 days?

```
SELECT
    c.customer_id,
    c.customer_name,
    MAX(o.order_date) AS last_order
FROM
    customers AS c
    LEFT JOIN
        orders AS o ON c.customer_id = o.customer_id
GROUP BY c.customer_id , c.customer_name
HAVING MAX(o.order_date) < '2026-04-30'
      OR MAX(o.order_date) IS NULL
ORDER BY last_order;
```

	customer_id	customer_name	last_order
▶	4	Vivek Chopra	NULL
	18	Nisha Joshi	NULL
	32	Nikhil Chauhan	NULL
	49	Kavita Bansal	NULL
	54	Gaurav Joshi	NULL
	58	Ritu Agarwal	NULL
	60	Gaurav Yadav	NULL
	63	Divya Joshi	NULL
	84	Neha Kapoor	NULL
	87	Deepak Nair	NULL
	12	Aman Kumar	2026-03-21

14

What is the best-selling category?

```
SELECT
    p.category, SUM(oi.quantity) AS unit_sold
FROM
    products AS p
    JOIN
        order_items AS oi ON oi.product_id = p.product_id
GROUP BY p.category
ORDER BY unit_sold DESC;
```

	category	unit_sold
▶	Electronics	3874
	Clothing	2951
	Home & Kitchen	2408
	Sports	1677
	Beauty	1468
	Books	1252
	Grocery	1205

15

Which product generates the highest revenue?

```
SELECT
    p.product_name,
    ROUND(SUM(oi.quantity * oi.unit_price), 2) AS rev
FROM
    products AS p
    JOIN
    order_items oi ON oi.product_id = p.product_id
GROUP BY p.product_name
ORDER BY rev DESC
LIMIT 1;
```

	product_name	rev
▶	Smartphone X12	9005526.00

16

Which product generates the lowest revenue?

```
SELECT
    p.product_name,
    ROUND(SUM(oi.quantity * oi.unit_price), 2) AS rev
FROM
    products AS p
    JOIN
    order_items oi ON oi.product_id = p.product_id
GROUP BY p.product_name
ORDER BY rev ASC
LIMIT 1;
```

	product_name	rev
▶	Self-Help Bestseller	135761.00

17

On average, how many products are bought per order?

```
SELECT
    ROUND(AVG(product_count), 2) AS avg_products_per_order
FROM
    (SELECT
        order_id, SUM(quantity) AS product_count
    FROM
        order_items
    GROUP BY order_id) t;
```

	avg_products_per_order
▶	7.42

18

Which orders are worth more than 10,000?

```
SELECT
    order_id,
    ROUND(SUM(quantity * unit_price), 2) AS order_value
FROM
    order_items
GROUP BY order_id
HAVING SUM(quantity * unit_price) > 10000
ORDER BY order_value DESC
LIMIT 10;
```

	order_id	order_value
▶	891	176982.00
	1952	172283.00
	762	150489.00
	1842	144687.00
	1373	135991.00
	311	133988.00
	1581	129583.00
	1343	126988.00
	304	123988.00
	39	113688.00

19

Which customers have the highest average order value?

```
SELECT
  customer_id, ROUND(AVG(order_value), 2) AS avg_order_value
FROM
  (SELECT
    o.customer_id,
    o.order_id,
    SUM(oi.quantity * oi.unit_price) AS order_value
  FROM
    orders AS o
  JOIN order_items AS oi ON o.order_id = oi.order_id
  GROUP BY o.customer_id , o.order_id) t
GROUP BY customer_id
ORDER BY avg_order_value DESC
LIMIT 5;
```

	customer_id	avg_order_value
▶	79	39412.80
	45	36465.65
	76	36202.00
	72	32881.75
	59	32559.91

20

What are the top 3 cities by revenue?

```
SELECT
  c.city, ROUND(SUM(oi.quantity * oi.unit_price), 2) AS rev
FROM
  orders AS o
  JOIN
  customers AS c ON o.customer_id = c.customer_id
  JOIN
  order_items AS oi ON oi.order_id = o.order_id
GROUP BY c.city
ORDER BY rev DESC
LIMIT 3;
```

	city	rev
▶	Delhi	5435283.00
	Bengaluru	5006267.00
	Kolkata	4635728.00

21

How are customers ranked based on revenue?

```
with customer_rev as (
  select o.customer_id, sum(oi.quantity*oi.unit_price) as rev
  from orders as o
  join order_items as oi on oi.order_id = o.order_id
  group by o.customer_id
)
select customer_id, round(rev,2) as revenue,
       Dense_rank() over(order by rev desc) as rev_rank
  from customer_rev
order by rev_rank
limit 10;
```

	customer_id	revenue	rev_rank
▶	13	1673772.00	1
	30	1529530.00	2
	46	1500017.00	3
	77	1338409.00	4
	25	1322346.00	5
	68	1285761.00	6
	74	1280445.00	7
	43	1145022.00	8
	41	1102554.00	9
	24	1085611.00	10

22

What is the top-selling product in each category?

```
with product_sales as (
  select p.category, p.product_name, sum(oi.quantity) as total_sold
  from products p
  join order_items oi on oi.product_id = p.product_id
  group by p.category, p.product_name)

select category, product_name, total_sold
  from (
    select * , row_number() over (partition by category order by total_sold desc)
      as rnk
  from product_sales) t
 where rnk = 1;
```

	category	product_name	total_sold
▶	Beauty	Hair Dryer	399
	Books	Fiction Novel Pack	458
	Clothing	Casual Sneakers	406
	Electronics	Smartphone X12	474
	Grocery	Organic Rice 5kg	433
	Home & Kitchen	Air Fryer	429
	Sports	Badminton Racket	457

23

How has our revenue accumulated over time — running total?

```
with daily_sales as (
  select o.order_date, sum(oi.quantity*oi.unit_price) as daily_revenue
  from orders o
  join order_items oi on o.order_id = oi.order_id
  group by o.order_date)

select order_date, daily_revenue,
       sum(daily_revenue) over (order by order_date) as running_rev
from daily_sales
order by order_date
limit 15;
```

	order_date	daily_revenue	running_rev
▶	2025-01-01	39553815.00	39553815.00
	2025-01-03	79107630.00	118661445.00
	2025-01-04	79107630.00	197769075.00
	2025-01-05	39553815.00	237322890.00
	2025-01-06	118661445.00	355984335.00
	2025-01-07	39553815.00	395538150.00
	2025-01-08	39553815.00	435091965.00
	2025-01-09	39553815.00	474645780.00
	2025-01-10	39553815.00	514199595.00

24

When was each customer's previous order placed?

```
select customer_id, order_id, order_date,
       lag(order_date) over (partition by customer_id order by order_date) as Previous_order_date
from orders
order by customer_id, order_date
limit 20;
```

	customer_id	order_id	order_date	Previous_order_date
▶	1	295	2025-01-27	NULL
	1	1611	2025-01-29	2025-01-27
	1	163	2025-02-23	2025-01-29
	1	1150	2025-03-05	2025-02-23
	1	1861	2025-03-29	2025-03-05
	1	222	2025-04-01	2025-03-29
	1	1522	2025-06-08	2025-04-01
	1	210	2025-08-29	2025-06-08
	1	1063	2025-09-14	2025-08-29
	1	689	2025-09-20	2025-09-14

25

When was each customer's next order placed, if any?

```
select customer_id, order_id, order_date,
       lead(order_date) over (partition by customer_id order by order_date) as next_order_date
from orders
order by customer_id, order_date
limit 20;
```

	customer_id	order_id	order_date	next_order_date
▶	1	295	2025-01-27	2025-01-29
	1	1611	2025-01-29	2025-02-23
	1	163	2025-02-23	2025-03-05
	1	1150	2025-03-05	2025-03-29
	1	1861	2025-03-29	2025-04-01
	1	222	2025-04-01	2025-06-08
	1	1522	2025-06-08	2025-08-29
	1	210	2025-08-29	2025-09-14
	1	1063	2025-09-14	2025-09-20
	1	689	2025-09-20	2025-11-08

26

How many days are there between a customer's consecutive orders?

```
select customer_id, order_date,
       datediff(order_date, lag(order_date) over(partition by customer_id order
       by order_date)) as days_between_orders
from orders
order by customer_id, order_date
limit 10;
```

	customer_id	order_date	days_between_orders
▶	1	2025-01-27	NULL
	1	2025-01-29	2
	1	2025-02-23	25
	1	2025-03-05	10
	1	2025-03-29	24
	1	2025-04-01	3
	1	2025-06-08	68
	1	2025-08-29	82
	1	2025-09-14	16
	1	2025-09-20	6

27

Who are the top 3 customers by revenue?

```
with customer_rev as (
  select o.customer_id, sum(oi.quantity*oi.unit_price) as revenue
  from orders o
  join order_items oi on o.order_id = oi.order_id
  group by o.customer_id
)

select customer_id, round(revenue,2) as rev, rnk
from (
  select * , Dense_rank() over(order by revenue desc) as rnk
  from customer_rev) t
where rnk<=3;
```

	customer_id	rev	rnk
▶	13	1673772.00	1
	30	1529530.00	2
	46	1500017.00	3

28

What percentage does each product contribute to total revenue?

```
with product_revenue as (
  select p.product_name, sum(oi.quantity*oi.unit_price) as revenue
  from products p
  join order_items oi on p.product_id = oi.product_id
  group by p.product_name
)

select product_name, revenue,
round(revenue * 100 / sum(revenue) over (),2) as revenue_percentage
from product_revenue
order by revenue desc
limit 10;
```

	product_name	revenue	revenue_percentage
▶	Smartphone X12	9005526.00	22.77
	Home Theater System	6335604.00	16.02
	4K Action Camera	3707588.00	9.37
	Air Fryer	1930071.00	4.88
	Smartwatch	1839632.00	4.65
	Mechanical Keyboard	1371608.00	3.47
	Mixer Grinder	1066419.00	2.70
	Winter Jacket	1040653.00	2.63
	Non-Stick Cookware Set	965580.00	2.44
	Bluetooth Speaker	834666.00	2.11

29

What is the month-over-month revenue growth?

```
with monthly_sales as (
  select date_format(o.order_date, '%Y-%M') as month,
         sum(oi.quantity*oi.unit_price) as revenue
  from orders o
  join order_items oi on oi.order_id = o.order_id
  group by month
)

select month, revenue,
       round(lag(revenue) over(order by month), 2) previous_month_revenue,
       round(100* (revenue - LAG(revenue) over (order by month)) / lag(revenue)
         over (order by month),2) as growth_percentage
from monthly_sales
order by month;
```

	month	revenue	previous_month_revenue	growth_percentage
▶	2025-April	967341.00	NULL	NULL
	2025-August	980876.00	967341.00	1.40
	2025-December	2734547.00	980876.00	178.79
	2025-February	727020.00	2734547.00	-73.41
	2025-January	857038.00	727020.00	17.88
	2025-July	1243632.00	857038.00	45.11
	2025-June	1415257.00	1243632.00	13.80
	2025-March	945215.00	1415257.00	-33.21
	2025-May	740314.00	945215.00	-21.68
	2025-November	2424781.00	740314.00	227.53

30

What was each customer's highest-value order?

```
with order_values as (
  select o.customer_id, o.order_id, sum(oi.quantity*oi.unit_price) as order_value
  from orders o
  join order_items oi on oi.order_id = o.order_id
  group by o.customer_id, o.order_id
)

select customer_id, order_id, order_value
from (
  select *, Row_number() over(partition by customer_id order by order_value desc)
  as rnk
  from order_values
) t
where rnk = 1
order by order_value desc
limit 10;
```

	customer_id	order_id	order_value
▶	72	891	176982.00
	71	1952	172283.00
	30	762	150489.00
	13	1842	144687.00
	45	1373	135991.00
	59	311	133988.00
	14	1581	129583.00
	47	304	123988.00
	77	39	113688.00
	80	289	112485.00

31

What is each customer's lifetime value — CLV?

```
SELECT
  o.customer_id,
  SUM(oi.quantity * oi.unit_price) AS customer_lifetime_value
FROM
  orders o
  JOIN
  order_items oi ON oi.order_id = o.order_id
GROUP BY o.customer_id
ORDER BY customer_lifetime_value DESC
LIMIT 10;
```

	customer_id	customer_lifetime_value
▶	13	1673772.00
	30	1529530.00
	46	1500017.00
	77	1338409.00
	25	1322346.00
	68	1285761.00
	74	1280445.00
	43	1145022.00
	41	1102554.00
	24	1085611.00

32

What percentage of customers make repeat purchases?

```
with customer_orders as (
  select customer_id, count(*) as total_orders
  from orders
  group by customer_id
)

select round(100.0 * count(case when total_orders > 1 then 1 end) / count(*), 2)
as repeat_purchase_rate
from customer_orders;
```

	repeat_purchase_rate
▶	100.00

33

How many customers are new vs returning?

```
with first_order as (
  select customer_id, min(order_date) as first_order_date
  from orders
  group by customer_id
)

select
  case when o.order_date = f.first_order_date then 'New customer'
  else 'Returning customer' end as customer_type,
  count(*) as total_orders
from orders as o
join first_order as f on o.customer_id = f.customer_id
group by customer_type;
```

	customer_type	total_orders
▶	Returning customer	1917
	New customer	83

34

How many customers were active each month — retention?

```
with monthly_orders as (
  select distinct customer_id, date_format(order_date, '%Y-%m') as order_month
  from orders
)

select order_month, count(distinct customer_id) as active_customers
from monthly_orders
group by order_month
order by order_month;
```

	order_month	active_customers
▶	2025-01	19
	2025-02	18
	2025-03	23
	2025-04	23
	2025-05	21
	2025-06	30
	2025-07	28
	2025-08	32
	2025-09	31
	2025-10	42
	2025-11	44

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Which customers purchase from more than one category?

```
SELECT
  o.customer_id,
  COUNT(DISTINCT p.category) AS categories_purchased
FROM
  orders AS o
  JOIN
  order_items AS oi ON o.order_id = oi.order_id
  JOIN
  products AS p ON oi.product_id = p.product_id
GROUP BY o.customer_id
HAVING COUNT(DISTINCT p.category) > 1
ORDER BY categories_purchased DESC;
```

	customer_id	categories_purchased
▶	1	7
	2	7
	3	7
	79	7
	80	7
	7	7
	8	7
	81	7
	10	7

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Which two products are most frequently purchased together?

```
SELECT
  oi1.product_id AS product_1,
  oi2.product_id AS product_2,
  COUNT(*) AS purchase_count
FROM
  order_items AS oi1
  JOIN
  order_items AS oi2 ON oi1.order_id = oi2.order_id
  AND oi1.product_id < oi2.product_id
GROUP BY oi1.product_id, oi2.product_id
ORDER BY purchase_count DESC
LIMIT 10;
```

	product_1	product_2	purchase_count
▶	9	34	15
	3	13	15
	12	36	14
	16	18	14
	2	35	14
	24	29	14
	6	28	14
	18	28	13
	7	8	13
	12	35	13

On average, how many days pass between a customer's orders?

```
SELECT
  customer_id,
  round(AVG(days_between_orders),2) AS avg_days_between_orders
FROM (
  SELECT
    customer_id,
    DATEDIFF(
      order_date,
      LAG(order_date) OVER (
        PARTITION BY customer_id
        ORDER BY order_date
      )
    ) AS days_between_orders
  FROM orders
) AS t
WHERE days_between_orders IS NOT NULL
GROUP BY customer_id;
```

	customer_id	avg_days_between_orders
▶	1	28.07
	2	3.04
	3	62.13
	5	23.17
	6	29.00
	7	24.11
	8	7.81
	9	4.11

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Which product category is growing the fastest?

```
with monthly_category_sales as (
  select date_format(o.order_date, '%y-%m') as month_name,
         p.category, sum(oi.quantity * oi.unit_price) as revenue
  from orders as o
  join order_items as oi on oi.order_id = o.order_id
  join products as p on p.product_id = oi.product_id
  group by month_name, p.category
)

select month_name, category, round(revenue,2) as rev,
       round(revenue - lag(revenue) over (partition by category order by month_name), 2)
       as revenue_growth
  from monthly_category_sales
 order by revenue_growth desc
 limit 15;
```

	month_name	category	rev	revenue_growth
▶	26-06	Electronics	3279210.00	948628.00
	26-03	Electronics	1656655.00	802728.00
	25-11	Electronics	1664279.00	564265.00
	25-06	Electronics	941168.00	559440.00
	26-05	Electronics	2330582.00	471294.00
	25-09	Electronics	805880.00	358093.00
	25-10	Electronics	1100014.00	294134.00
	26-06	Home & Kitchen	723259.00	290089.00
	25-12	Electronics	1903451.00	239172.00
	26-04	Electronics	1859288.00	202633.00
	26-03	Home & Kitchen	385222.00	161339.00

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Which customers are at risk of churn?

```
SELECT
  customer_id, MAX(order_date) AS last_order_date
FROM
  orders
GROUP BY customer_id
HAVING MAX(order_date) < '2026-04-30'
ORDER BY last_order_date
LIMIT 15;
```

	customer_id	last_order_date
▶	12	2026-03-21
	1	2026-03-24
	31	2026-04-02
	85	2026-04-15

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How do customers look based on RFM — Recency, Frequency, Monetary- analysis?

```

SELECT
  o.customer_id,
  DATEDIFF('2026-07-30', MAX(o.order_date)) AS recency_days,
  COUNT(DISTINCT o.order_id) AS frequency,
  SUM(oi.quantity * oi.unit_price) AS monetary
FROM
  orders AS o
  JOIN
  order_items AS oi ON o.order_id = oi.order_id
GROUP BY o.customer_id
ORDER BY monetary DESC
limit 15;

```

	customer_id	recency_days	frequency	monetary
▶	13	5	65	1673772.00
	30	9	77	1529530.00
	46	6	62	1500017.00
	77	6	64	1338409.00
	25	9	79	1322346.00
	68	10	65	1285761.00
	74	8	67	1280445.00
	43	5	61	1145022.00
	41	27	52	1102554.00
	24	10	53	1085611.00
	10	21	54	994589.00
	69	13	53	974254.00
	71	10	61	964473.00

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What is the daily sales trend?

```

SELECT
  o.order_date,
  SUM(oi.quantity * oi.unit_price) AS daily_revenue
FROM
  orders AS o
  JOIN
  order_items AS oi ON o.order_id = oi.order_id
GROUP BY o.order_date
ORDER BY o.order_date
LIMIT 15;

```

	order_date	daily_revenue
▶	2025-01-01	14282.00
	2025-01-03	79732.00
	2025-01-04	40491.00
	2025-01-05	27989.00
	2025-01-06	99370.00
	2025-01-07	11489.00
	2025-01-08	31289.00
	2025-01-09	13494.00
	2025-01-10	64188.00
	2025-01-11	18238.00

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What are the top 10 best-selling products?

```
SELECT
    p.product_name, SUM(oi.quantity) AS total_quantity
FROM
    order_items AS oi
    JOIN
        products AS p ON oi.product_id = p.product_id
GROUP BY p.product_name
ORDER BY total_quantity DESC
LIMIT 10;
```

	product_name	total_quantity
▶	Smartphone X12	474
	Fiction Novel Pack	458
	Badminton Racket	457
	Cricket Bat	450
	Organic Rice 5kg	433
	Air Fryer	429
	Table Lamp	427
	Cold-Pressed Oil 1L	427
	Non-Stick Cookware Set	420
	4K Action Camera	412

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Who are the top 10 customers by revenue?

```
SELECT
    c.customer_name,
    SUM(oi.quantity * oi.unit_price) AS total_revenue
FROM
    customers AS c
    JOIN
        orders AS o ON o.customer_id = c.customer_id
    JOIN
        order_items oi ON oi.order_id = o.order_id
GROUP BY c.customer_name
ORDER BY total_revenue DESC
LIMIT 10;
```

	customer_name	total_revenue
▶	Rahul Iyer	1673772.00
	Nikhil Mishra	1529530.00
	Riya Rana	1523531.00
	Simran Verma	1500017.00
	Manish Gupta	1449062.00
	Neha Sharma	1339444.00
	Anil Verma	1338409.00
	Komal Malhotra	1322346.00
	Pallavi Chauhan	1285761.00
	Rahul Rana	1222068.00

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How much revenue comes from male vs female customers?

```
SELECT
  c.gender, SUM(oi.quantity * oi.unit_price) AS revenue
FROM
  customers AS c
  JOIN
    orders AS o ON o.customer_id = c.customer_id
  JOIN
    order_items oi ON oi.order_id = o.order_id
GROUP BY c.gender
ORDER BY revenue DESC;
```

	gender	revenue
▶	Male	20859804.00
	Female	18694011.00

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What is our customer churn rate?

```
with customer_status as (
  select customer_id, max(order_date) as last_order_date,
  case when max(order_date) < '2026-04-30' then 1 else 0 end as is_churned
  from orders
  group by customer_id
)
select round(100* sum(is_churned) / count(*),2) as churn_rate_per
from customer_status;
```

	churn_rate_per
▶	5.00

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What percentage of orders get cancelled?

```
SELECT
    ROUND(100 * COUNT(CASE
        WHEN order_status = 'cancelled' THEN 1
    END) / COUNT(*),
    2) AS cancellation_rate
FROM
    orders;
```

	cancellation_rate
▶	12.00

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How many orders were delivered vs cancelled?

```
SELECT
    order_status, COUNT(*) AS total_orders
FROM
    orders
WHERE
    order_status IN ('Delivered' , 'Cancelled')
GROUP BY order_status;
```

	order_status	total_orders
▶	Delivered	1649
	Cancelled	240

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What is the worst-selling category?

```
SELECT
    p.category, SUM(oi.quantity) AS units_sold
FROM
    products AS p
    JOIN
    order_items oi ON p.product_id = oi.product_id
GROUP BY p.category
ORDER BY units_sold ASC
LIMIT 1;
```

	category	units_sold
▶	Grocery	1205

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What is the most expensive product sold?

```
SELECT
    p.product_name,
    MAX(oi.unit_price) AS expensive_prd
FROM
    products AS p
    JOIN
    order_items AS oi ON p.product_id = oi.product_id
GROUP BY p.product_name
ORDER BY expensive_prd DESC
LIMIT 1;
```

```
SELECT
    product_name, unit_price
FROM
    products
ORDER BY unit_price DESC
LIMIT 1;
```

	product_name	expensive_prd
▶	Smartphone X12	18999.00

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Which month had the highest revenue?

```
with monthly_revenue as (  
  select date_format(o.order_date, '%Y-%m') as month_data,  
         sum(oi.quantity*oi.unit_price) as revenue  
  from orders as o  
 join order_items as oi on o.order_id = oi.order_id  
 group by month_data  
)  
  
select month_data, revenue  
from monthly_revenue  
order by revenue desc  
limit 1;
```

	month_data	revenue
▶	2026-06	5171310.00

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How many new customers were acquired each month?

```
with first_orders as (  
  select customer_id, min(order_date) as first_order_date  
  from orders  
 group by customer_id  
)  
  
select date_format(first_order_date, '%Y-%m') as acquisition_month, count(customer_id)  
      as new_customers  
from first_orders  
group by acquisition_month  
order by acquisition_month;
```

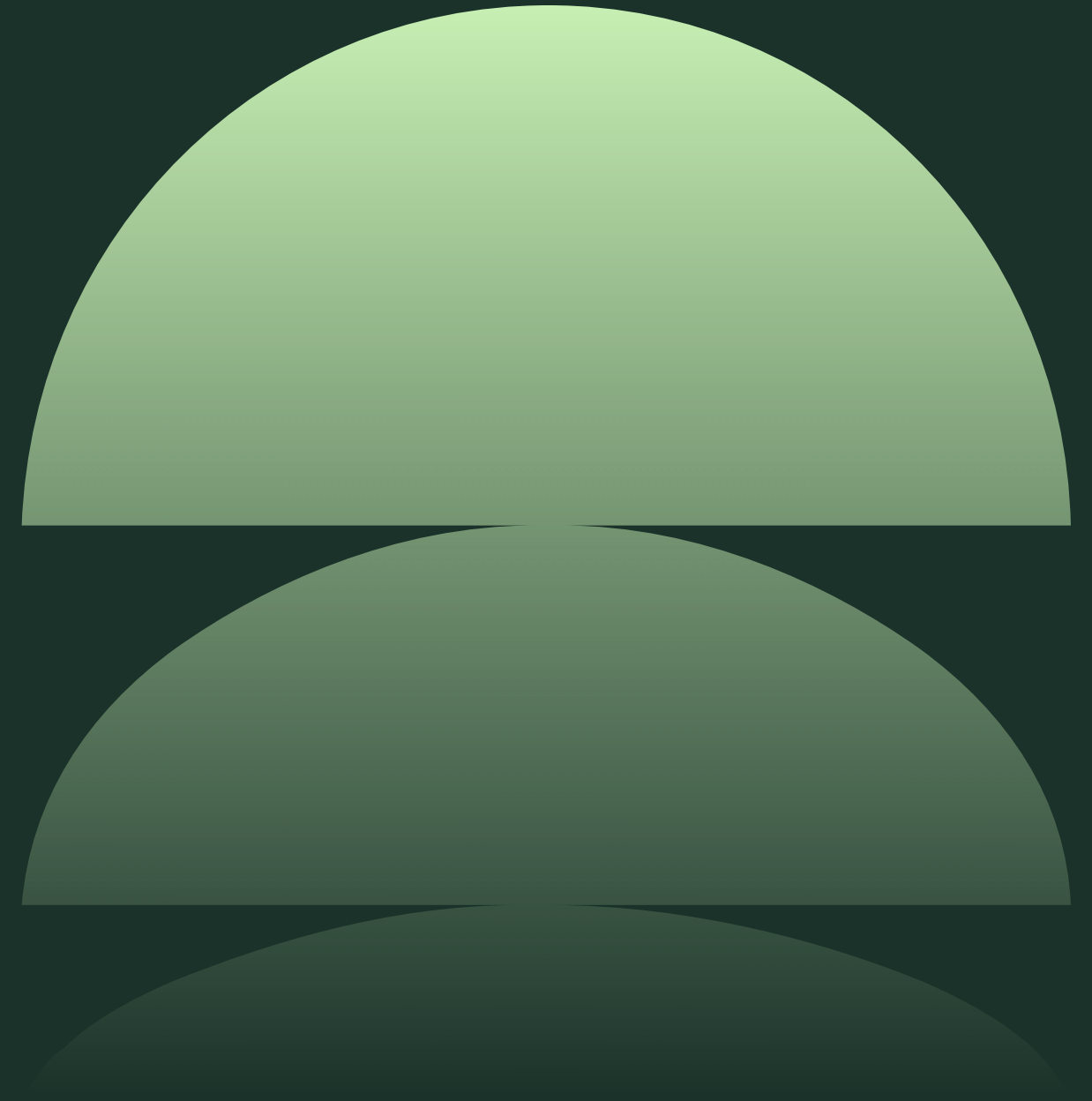
	acquisition_month	new_customers
▶	2025-01	19
	2025-02	6
	2025-03	5
	2025-04	1
	2025-05	5
	2025-06	4
	2025-07	5
	2025-08	3
	2025-09	2
	2025-10	4

What percentage does each category contribute to total revenue?

```
with category_revenue as (  
    select p.category, sum(oi.quantity * oi.unit_price) as revenue  
    from products as p  
    join order_items as oi on p.product_id = oi.product_id  
    group by p.category  
)  
  
select category, round(revenue, 2) as rev,  
    round(100* revenue / sum(revenue) over(), 2) as rev_contribution_perc  
from category_revenue  
order by rev desc;
```

	category	rev	rev_contribution_perc
▶	Electronics	25223126.00	63.77
	Home & Kitchen	5080392.00	12.84
	Clothing	4080449.00	10.32
	Sports	2064623.00	5.22
	Beauty	1829532.00	4.63
	Grocery	683745.00	1.73
	Books	591948.00	1.50

THANK YOU



SEPTEMBER

BY; NAKUL TYAGI
